

Week-4

- a) Implement FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

```
import pandas as pd

import numpy as np

df=pd.read_csv("C://Users//Admin//Downloads//6646ML//enjoysport.csv")

print(df)
```

O/P:

	Sky	Airtemp	Humid	Wind	Water	Forecast	Outcome
0	sunny	warm	normal	strong	warm	same	yes
1	sunny	warm	high	strong	warm	same	yes
2	rainy	cold	high	strong	warm	change	no
3	sunny	warm	high	weak	cool	change	yes

```
features=np.array(df)[:,-1]

print("features: ",features)
```

O/P: Features : [['sunny' 'warm' 'normal' 'strong' 'warm' 'same']
['sunny' 'warm' 'high' 'strong' 'warm' 'same']
['rainy' 'cold' 'high' 'strong' 'warm' 'change']
['sunny' 'warm' 'high' 'weak' 'cool' 'change']]

```
target=np.array(df)[:,-1]

print("Target:",target)
```

Target: ['yes' 'yes' 'no' 'yes']

```
for i,val in enumerate(target):
    if val=="yes":
        hypothesis=features[i].copy()
        break
print(hypothesis)
```

O/P: ['sunny' 'warm' 'normal' 'strong' 'warm' 'same']

```
for i,val in enumerate(features):  
    if target[i]=="yes":  
        for x in range(len(hypothesis)):  
            if hypothesis[x]!=val[x]:  
                hypothesis[x]="?"  
print(hypothesis)
```

O/P:

```
['sunny' 'warm' '?' '?' '?' '?']
```